C++

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Things to start with:

- C++ is compiled language:
  - sourcecode.cpp -> COMPILER -> executable.exe
  - Resulting .exe file in target machines native language

- Supports different programming paradigms:
  - procedural, generic and object-oriented programming

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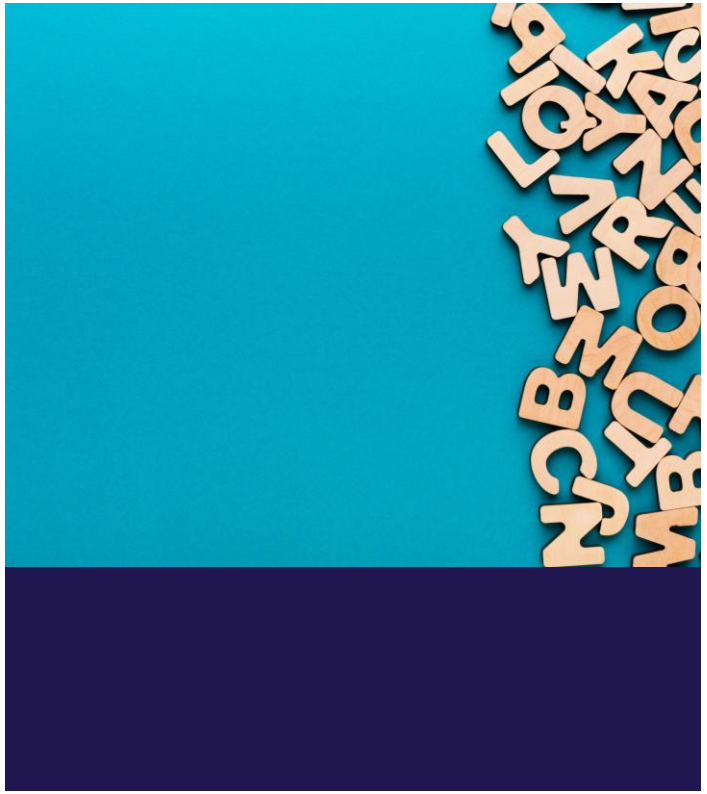
Variable types

- Integer type: int
  - Whole number value like 128
- Floating point type: double
  - Real value like 3.14
- Character type: char
  - One byte character like ‘a’

Variable types

- String type: string (actually std::string)
- String is not basic type in C++. It’s from standard library



Arrays

- Array for integer values:
  - int values[5]; // Array length 5
  - Indexes start from zero. Above valid indexes are 0,1,2,3,4
  - Compiler does not check if read/written out of bounds
    - Out of bounds causes runtime error

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Program structures

- Function name:
  - void myFunctionName() {}
- Function return value:
  - int myFunctionName() {}

Program structures

- Function arguments, parameters and return value

```
int addfunction(int a, int b) { return a+b;}
```

15 7 8

```
c = addfunction(7, 8); // function call with arguments
```

Program structures

- Code blocks are defined by curly brackets {}
- Variable is only visible on code block where variable declaration is made and in all code blocks that are inside that block
- Every code line inside code block ends with semicolon (;)

Program structures: keywords

- For loop to repeat code block so many times as wanted
  - for(int i = 0; i < 5; i++) {}
  - Variable “i” can be declared inside for statement
  - In above line i starting from zero, iterated as long as less than five, i added by one for each round
  - i can be initialized to selected value (above i = 0)
  - i can be added by what ever wanted: i += 2 (i = i + 2)

Function declaration

- If we want to use function in C++ code and implementation for the function is given later/somewhere else in the code, we must make function **declaration**
- Declaration informs the compiler about the existence of the function
- It makes possible for compiler to check parameter correctness in function call

Standard library operations

- Taken into use by include e.g.:
  - #include <iostream>
- std::cout for output stream
- std::cin for input stream
- Stream operators << and >> to handle streams
  - std::cout << “Show this text” << std::endl;

Standard library operations

- ofstream for file output
  - Needs to be included: #include<fstream>
  - Needs to be declared: std::ofstream myFile;
  - Needs to be opened: myFile.open(“filename.txt”);
  - Stream operator can be used: myFile << “save this text\n”;
  - Needs to be closed: myFile.close();

Standard library operations

- We have to use std:: before standard library operations to tell compiler in which namespace operation is located
  - std::cout
- Alternative way is to add: using namespace std; in beginning of the cpp file
  - std:: can be left out

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